

House Price Prediction – Summary

Objective: The objective of this project was to develop a machine learning model capable of predicting house prices based on various property characteristics. Accurate house price prediction can help buyers, sellers, and real estate businesses make informed decisions by estimating the fair market value of a property. The project focused on identifying the most influential factors affecting house prices and comparing the performance of different regression models.

Methodology: The Housing dataset was loaded and explored using the Pandas library. Initial data analysis included examining the dataset structure, identifying the target variable, and checking for missing values and duplicate records. Since the dataset contained categorical attributes such as air conditioning, furnishing status, and preferred area, these variables were converted into numerical form using one-hot encoding. The cleaned dataset was then divided into training and testing sets using an 80:20 ratio.

Two machine learning models were trained for price prediction: Linear Regression and Random Forest Regressor. Their performances were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. Additionally, data visualizations such as a price distribution histogram, correlation heatmap, and actual versus predicted price scatter plot were created to better understand the relationships within the data.

Results and Findings: The Linear Regression model achieved an MAE of 970,043.40, an RMSE of 1,324,506.96, and an R^2 score of 0.653. The Random Forest Regressor achieved an MAE of 1,021,546.04, an RMSE of 1,400,565.97, and an R^2 score of 0.612. Based on these evaluation metrics, the Linear Regression model performed better on this dataset by producing lower prediction errors and a higher R^2 score.

Feature importance analysis revealed that the area of the house was the most significant factor influencing house prices, accounting for approximately 46.84% of the prediction importance. Other important factors included the number of bathrooms, availability of air conditioning, parking facilities, and the number of stories in the property. These findings indicate that both the physical size of a house and its amenities play an important role in determining its market value.

Conclusion: The project successfully demonstrated the use of machine learning techniques for house price prediction. The results showed that Linear Regression was more suitable for this dataset than Random Forest Regressor. An interesting observation was that amenities such as air conditioning and parking had a considerable influence on house prices, highlighting that buyers value convenience and comfort in addition to property size. Based on these findings, real estate businesses should emphasize larger properties with desirable amenities when pricing and marketing houses. Overall, the developed model provides a useful tool for estimating property values and supporting data-driven decision-making in the real estate industry.